CAD for VLSI

Clock and Power Routings

Routing of Clock and Power Nets

- Different from other signal nets, clock and power are special routing problems
  - For clock nets, need to consider clock skew as well as delay
  - For power nets, need to consider current density (IR drop)
- => specialized routers for these nets
- Automatic tools for ASICs
- Often manually routed and optimized for microprocessors, with help from automatic tools

Clock Introduction

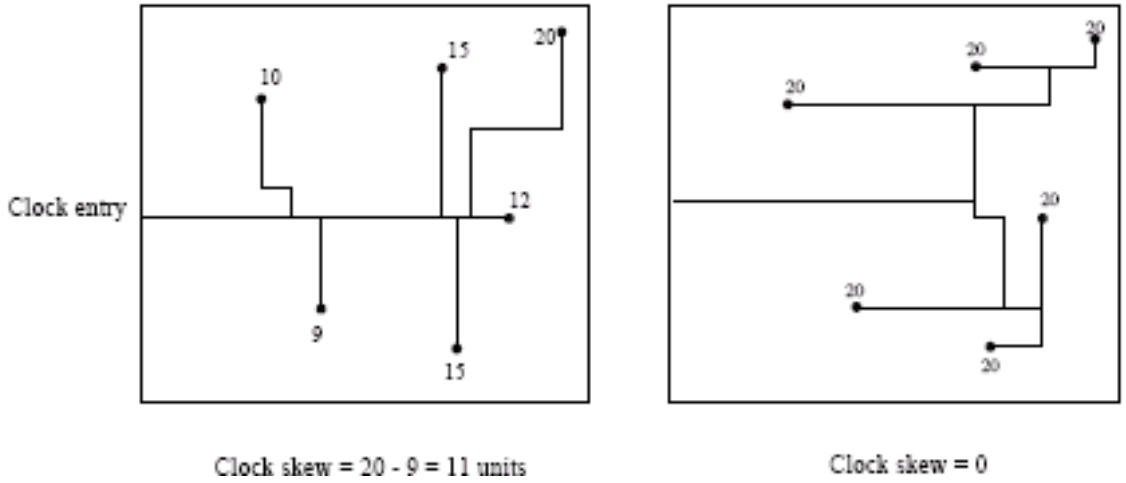
- For synchronized designs, data transfer between functional elements are synchronized by clock signals
- Clock signal are generated externally (ex: by PLL)
- Clock period equation

$$clock\ period \geq t_d + t_{skew} + t_{su} + t_{ds}$$

t_d:	Longest path through combinational logic
t_{skew}:	Clock skew
t_{su}:	Setup time of the synchronizing elements
t_{ds}:	Propagation delay within the synchronizing element

Clock Skew

- Clock skew is the maximum difference in the arrival time of a clock signal at two different components
- Clock skew forces designers to use a large time period between clock pulses
  - This makes the system slower
- Hence, in addition to other objectives, clock skew should be minimized during clock routing



Clock skew = 20 - 9 = 11 units

Clock skew = 0

Clock Design Problem

- What are the main concerns for clock design?
- Skew
  - No. 1 concern for clock networks
  - For increased clock frequency, skew may contribute over 10% of the system cycle time
- Power
  - Very important, as clock is a major power consumer!
  - It switches at every clock cycle!
- Noise
  - Clock is often a very strong aggressor
  - May need shielding
- Delay
  - Not really important
  - But slew rate is important (sharp transition)

The Clock Routing Problem

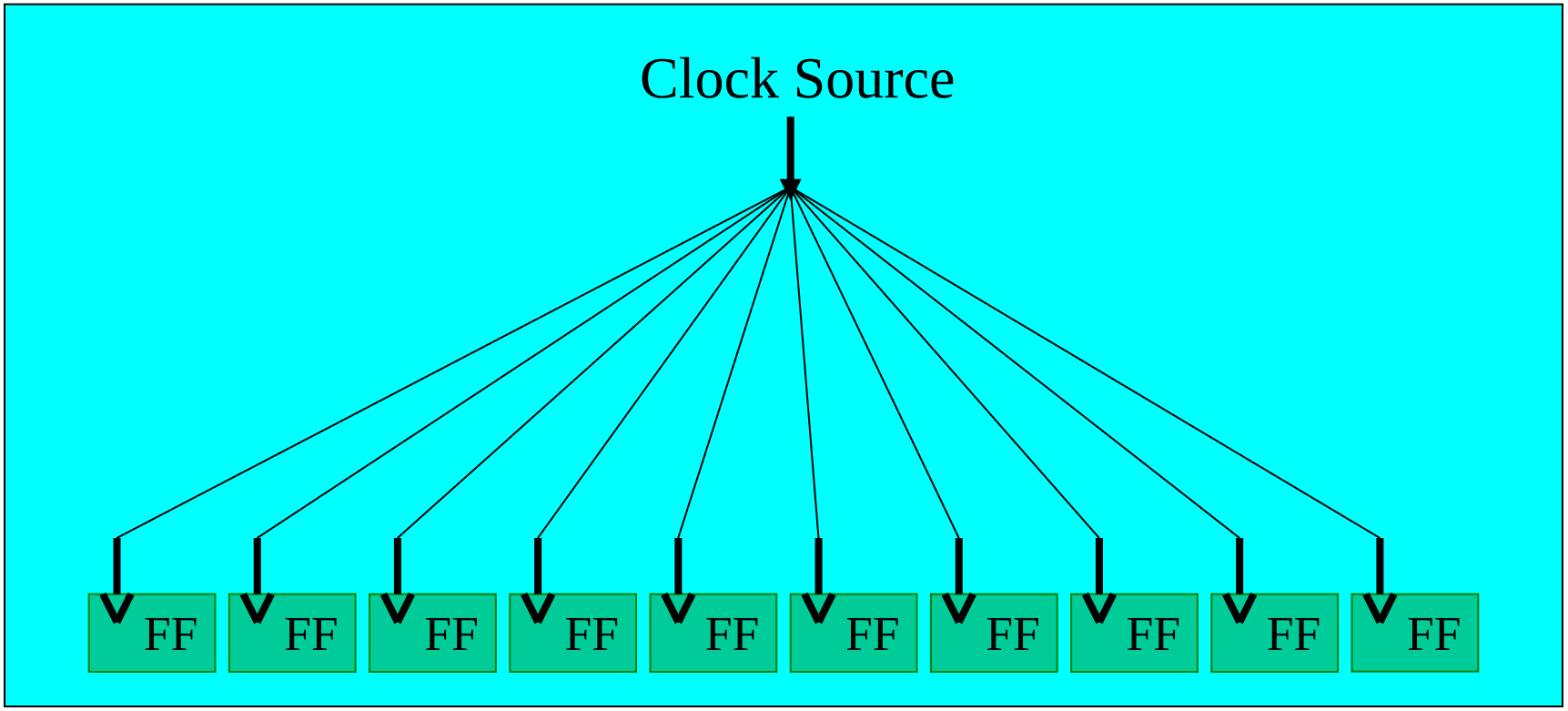
- Given a source and *n* sinks
- Connect all sinks to the source by an interconnect network (tree or non-tree) so as to minimize:
  - Clock Skew = $\max_{i,j} |t_i - t_j|$
  - Delay = $\max_i t_i$
  - Total wirelength
  - Noise and coupling effect

Clock Design Considerations

- Clock signal is global in nature, clock nets are usually very big
  - Significant interconnect capacitance and resistance
- So what are the techniques?
  - Routing
    - Clock tree versus clock mesh (non-tree or grid)
    - Balance skew and total wire length
  - Buffer insertion
    - Clock buffers to reduce clock skew, delay, and distortion in waveform
  - Wire sizing
    - To further tune the clock tree/mesh

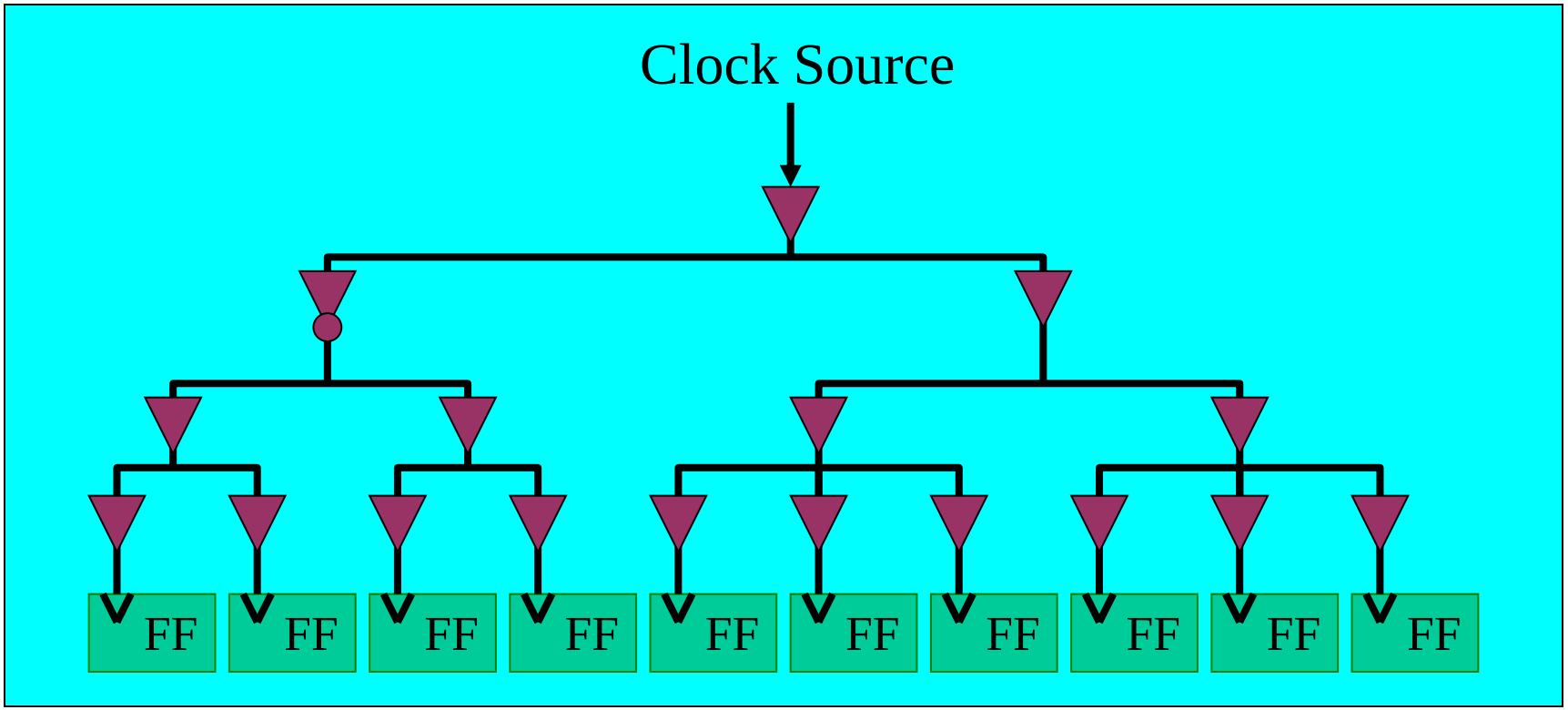
Clock Trees

- A path from the clock source to clock sinks

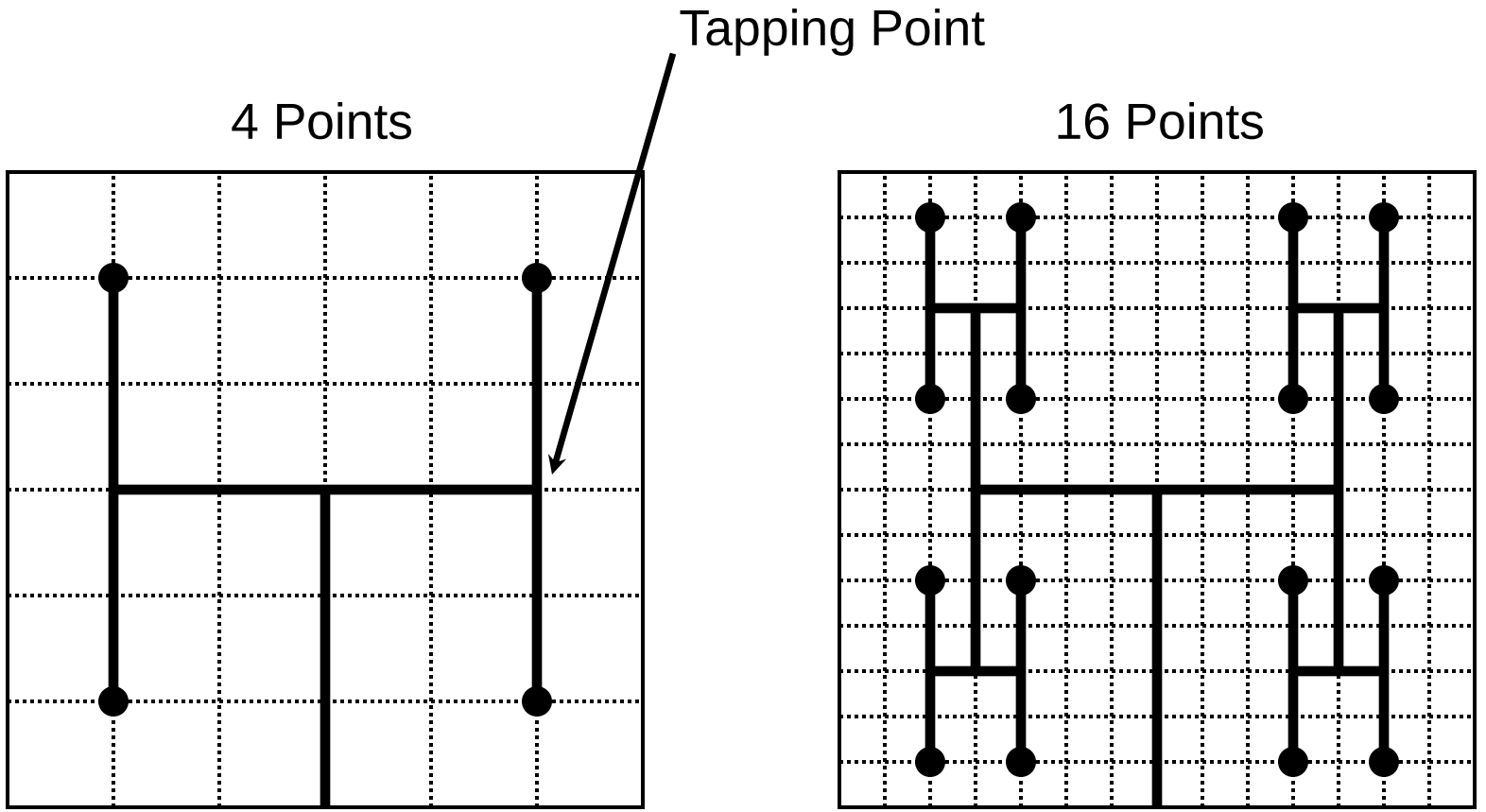


Clock Trees

- A path from the clock source to clock sinks



H-Tree Clock Routing



H-tree Algorithm

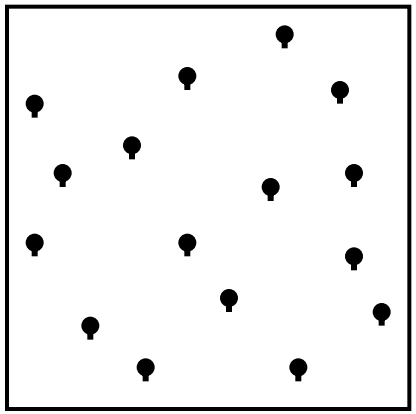
- Minimize skew by making interconnections to subunits equal in length
  - Regular pattern
  - The skew is 0 assuming delay is directly proportional to wirelength
    - Is this always the case???
- Can be used when terminals are evenly distributed
  - However, this is never the case in practice (due to blockage, and so on)
  - So strict (pure) H-trees are rarely used
  - However, still popular for top-level clock network design
  - Cons: too costly to be used everywhere

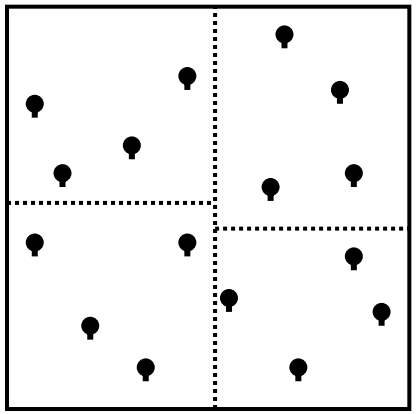
Can you think of another shape if non-rectilinear wires are allowed?

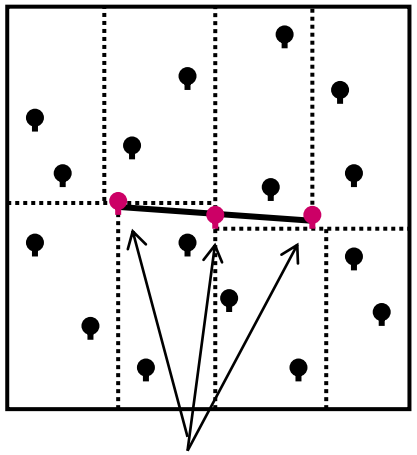
Method of Means and Medians (MMM)

- Applicable when the clock terminals are arbitrarily arranged
- Follows a strategy very similar to H-Tree
  - Recursively partition the terminals into two sets of equal size (median)
  - Then, connect the center of mass of the whole circuit to the centers of mass of the two sub-circuits (mean)
- Clock skew is only minimized heuristically
  - The resulting tree may not have zero-skew

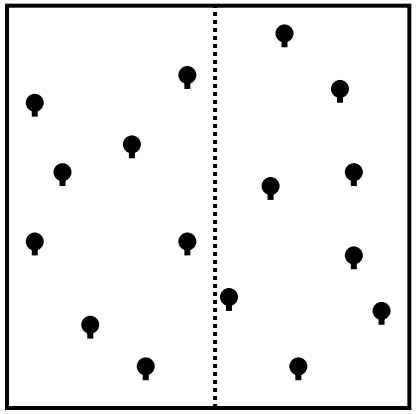
An Example of MMM

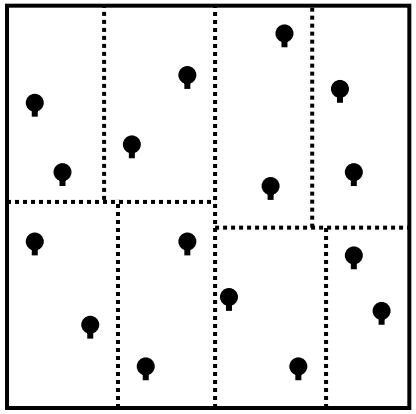


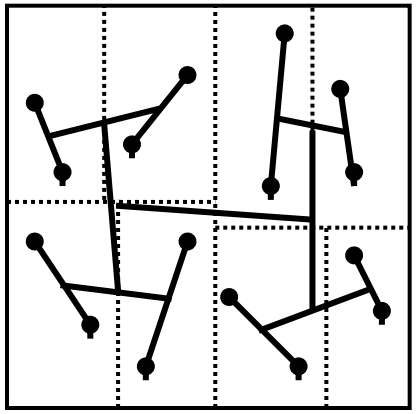




centers of mass



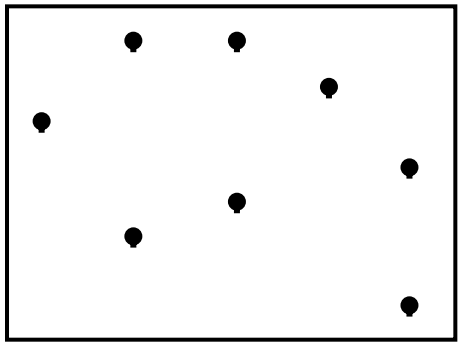


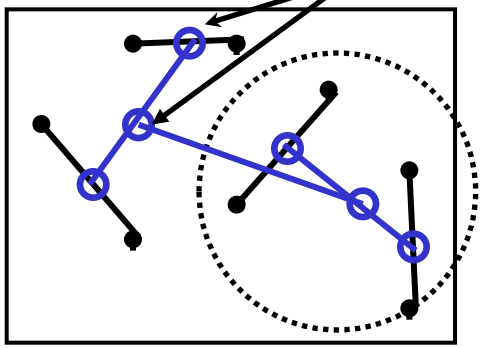


Geometric Matching Algorithm (GMA)

- MMM is a top-down algorithm, but GMA is a bottom-up algorithm
- Geometric matching of *n* endpoints:
  - Construct a set of *n*/2 line segments connecting *n* endpoints pairwise
  - No two line segments share an endpoint
  - The cost is the sum of the edge lengths
- The basic idea is to find a minimum cost geometric matching recursively
- Time complexity is $O(n^{2.5} \log n)$ for *n* endpoints

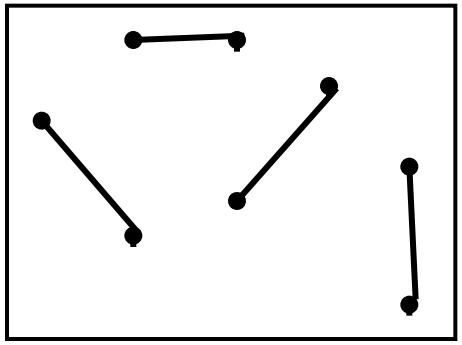
An Example of GMA



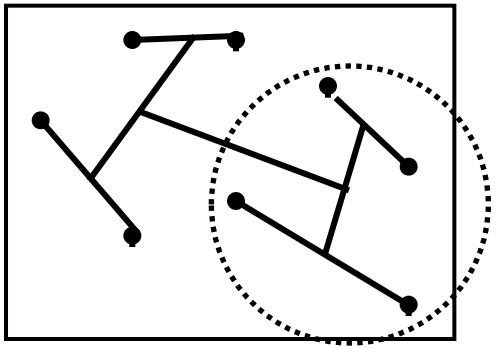


Tapping point
(not necessarily
the mid-point)

H-flipping



Apply geometric
matching recursively



Post-processing

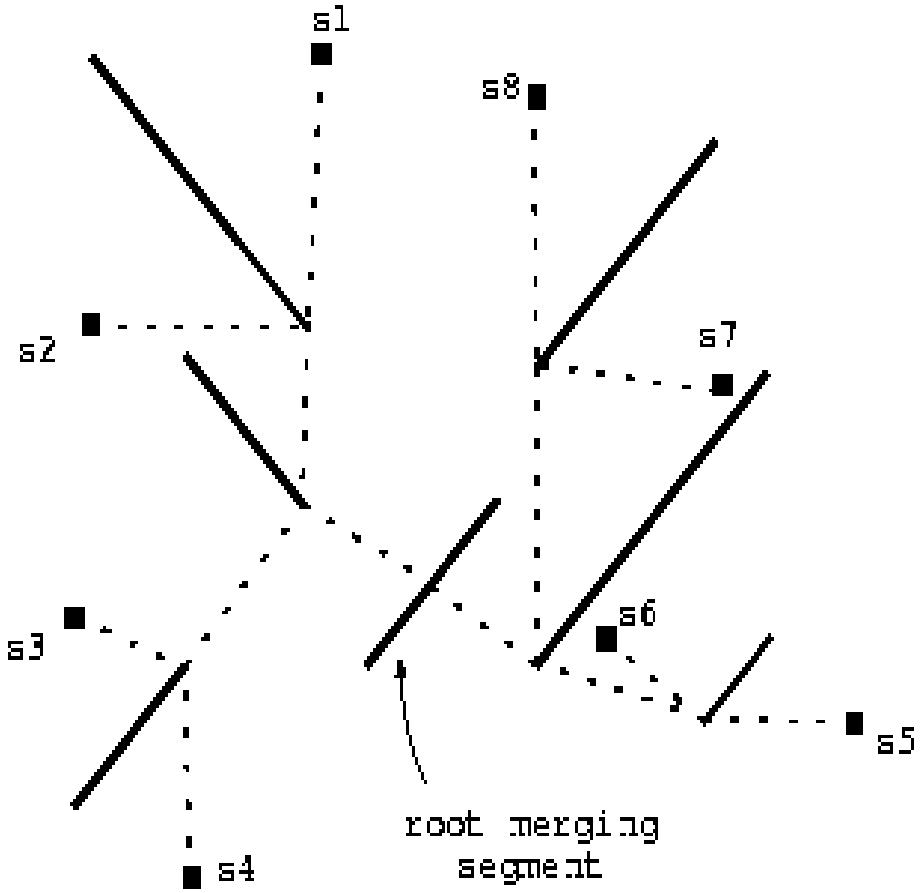
Can give zero skew clock tree

An Exact Zero Skew Algorithm

- ICCAD 1991 and TCAD 1993, Ren-Song Tsay
- A classic paper to manage clock skew
- Use Elmore delay model to compute delay
- Guarantee zero skew
  - Can easily to extended for zero skew or bounded skew
  - Can you think of a method to do it?
- Try to minimize wire length, but not done very well
  - Lots of follow up works to minimize total wire length while maintaining zero skew
  - DME and its extensions

Deferred Merge Embedding

- As its name implies, DME defers the merging as late as possible, to make sure minimal wire length cost for merging
- Independently proposed by several groups
  - Edahiro, NEC Res Dev, 1991
  - Chao et al, DAC'92
  - Boese and Kahng, ASIC'92
- DME needs an *abstract routing topology* as the input
- It has a bottom-up phase followed by a top-down process (sounds familiar?)

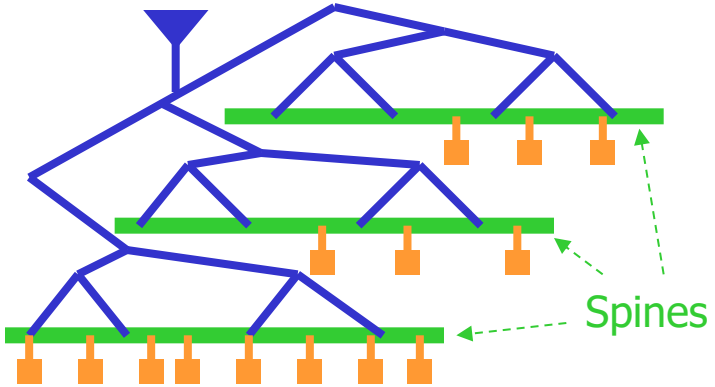


DME: An example of a tree of merging segments with sinks $s_1, \ldots, s_8$. The solid lines are merging segments and the dotted lines indicate edges between merging segments.

Some Thoughts/Trend

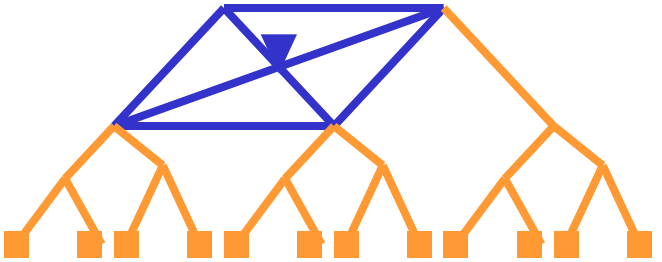
- Clock skew scheduling together with clock tree synthesis
  - Schedule the timing slack of a circuit to the individual registers for optimal performance and as a second criteria to increase the robustness of the implementation w.r.t. process variation
- Variability is a major nanometer concern
- Non-tree clock networks for variation-tolerance
  - How to analyze it?
  - The task is to investigate a combined optimization such that clock skew variability is reduced with minimum wirelength penalty

Non-tree: Spine & Mesh

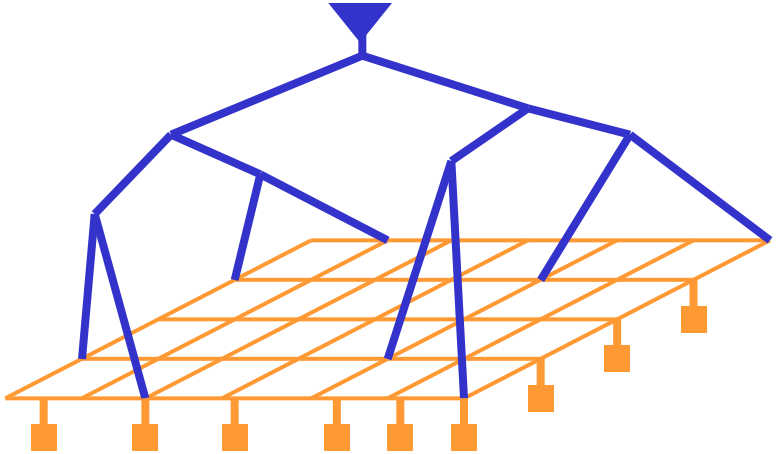


Applied in *Pentium* processor

[Kurd et. al. JSSC'01]



[Su et. al, ICCAD'01]



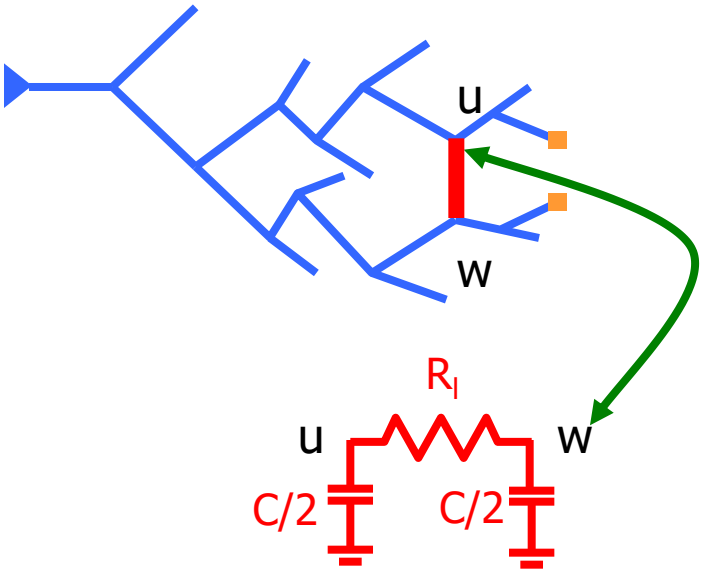
[Restle et. al, JSSC'01]

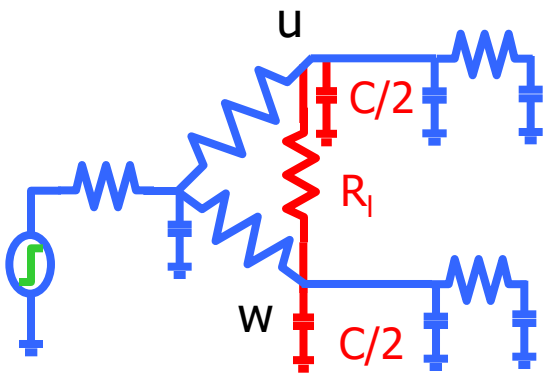
Applied in IBM microprocessor

- Very effective, huge wire

Non-tree: Link Perspective

- Non-tree = tree + links
- How to select link pairs is the key problem
- Link = link_capacitors + link_resistor
- Key issue: find the best links that can help the skew variation reduction the most!





[Rajaram et al, DAC'04]

Power Distribution/Routing

Power Distribution

- Power Distribution Network functions
  - Carry current from pads to transistors on chip
  - Maintain stable voltage with low noise
  - Provide average and peak power demands
  - Provide current return paths for signals
  - Avoid electromigration & self-heating wearout
  - Consume little chip area and wire
  - Easy to lay out

Power and Ground Routing

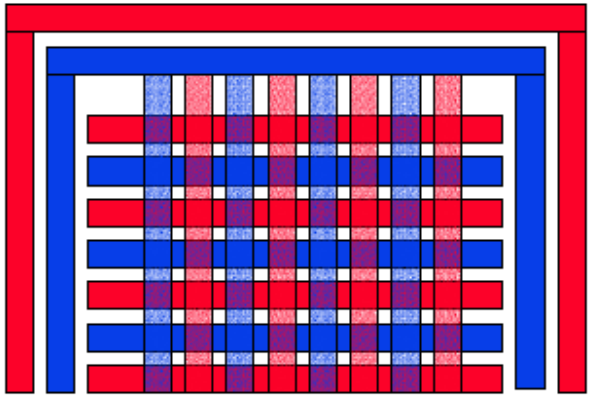
- Each standard cell or macro has power and ground signals, i.e., VDD (power) and GND (ground)
- They need to be connected as well
- You can imagine that they are HUGE NETWORKS!
- In general, P/G routings are pretty regular
- They have high priority as well
  - P/G routing resources are usually reserved
  - When you do global and detailed routing for signal nets, you cannot use up all the routing resources at each metal layers
    - Normally some design rules will be given (ex: 40% of top metal layers are reserved for P/G)

P/G Routing Main Objectives

- Routing resource
  - Need to balance the routing resource for P/G, clock and signals
- Voltage drop
  - Static (IR) and dynamic (L di/dt) voltage drops
  - More voltage drop means more gate delay
  - Usually less than 5-10% voltage drop is allowed
  - So you may need to size P/G wires accordingly
- Electrical migration
  - Too big current may cause EMI problem
- Others…

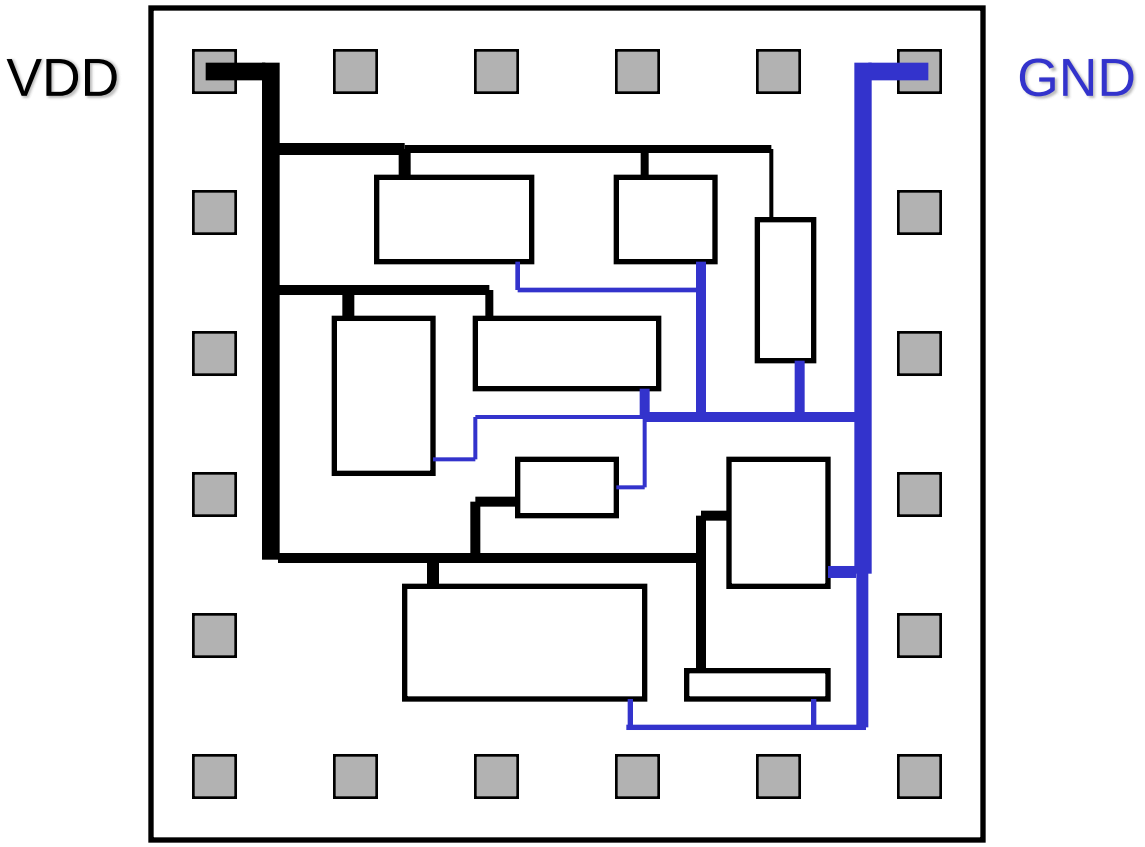
P/G Mesh (Grid Distribution)

- Power/Ground mesh will allow multiple paths from P/G sources to destinations
  - Less series resistance
  - Hierarchical power and ground meshes from upper metal layers to lower metal layers
    - All the way to M1 or M2 (stand cells)
  - Connection of lower layer layout/cells to the grid is through vias



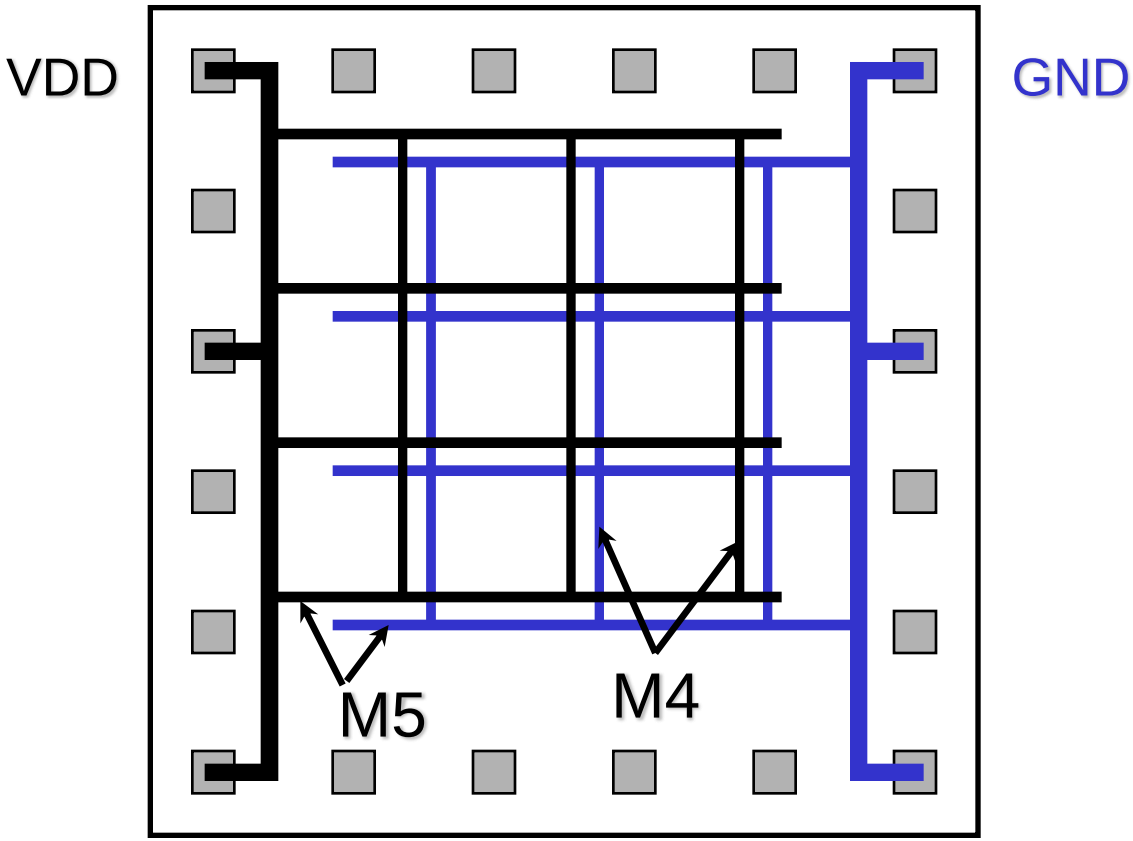
Using One Metal Layer

- One tree for VDD and another tree for GND



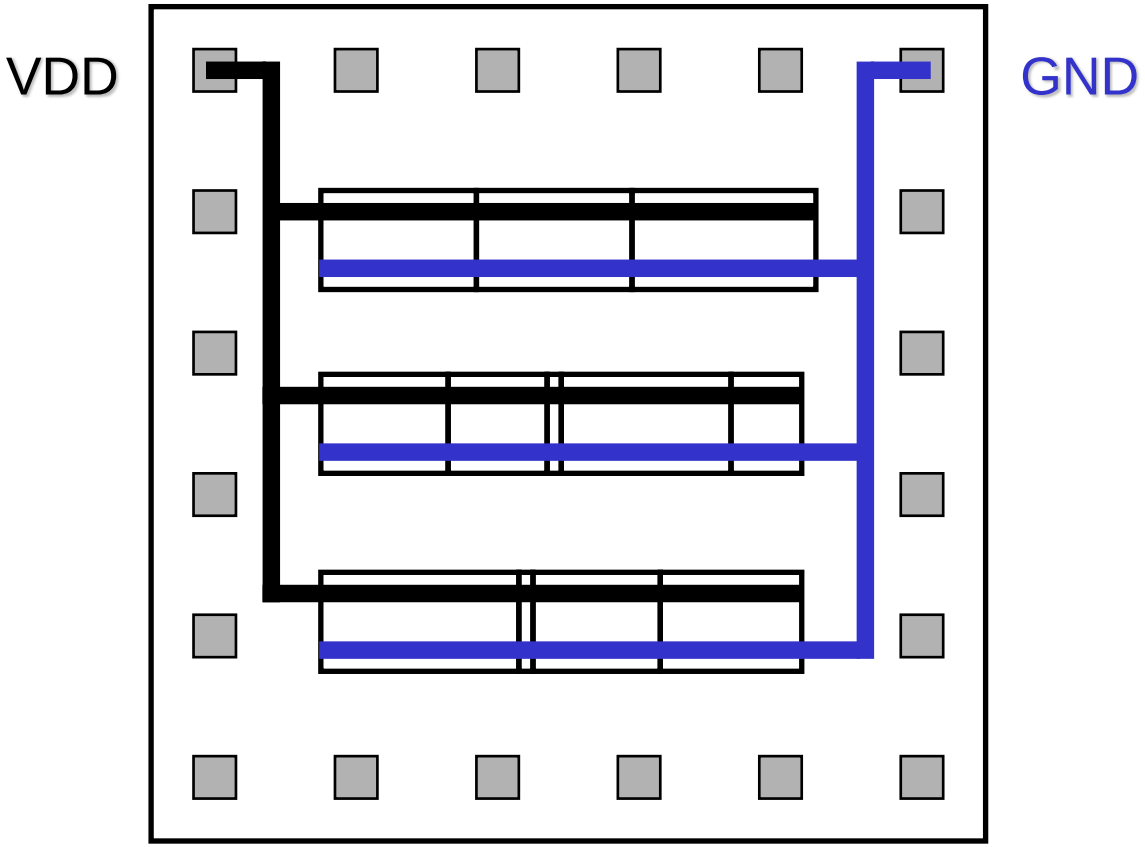
Using Two Metal Layers

- One 2D-grid for VDD and another one for GND



Gate Array & Standard Cell Design

- Inter-weaved combs



Some Thoughts/Trends

- P/G I/O pad co-optimization with classic physical design
- Decoupling capacitor can reduce P/G related voltage drop
  - Need to be planned together with floorplanning and placement
- Multiple voltage/frequency islands make the P/G problem and clock distributions more challenging

